

Intro
website:

Kevinqu2004.github.io/cs-584-physreal-computing

GitHub:

github.com/kevinqu2004/cs-584-physreal-computing



Main Drive

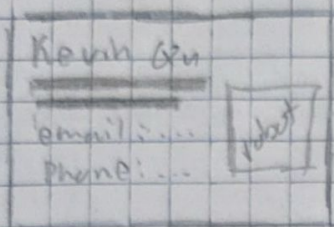
<https://drive.google.com/drive/folders/1DEhOzm10mPKIxZtTDUR6itkD8TfNk9hn?usp=sharing>



GitHub won't let me upload
videos/photos that are too big
so here's a google drive link

Business card

V1



name ✓
contact info ✓
interest ✓

based off another card I saw on Google images, the two lines kinda make it look more professional

V2

Saw a "card" in the shape of a model plane (it was 3D) and now I need one too



Ideas:

- make card out of aluminum
- make mini robot out of metal
- 3D print robot miniature
- cut out silhouette from two sides and glue them together

- still have to use card stock :/

- paper mechanism in 2D card (multi-layered)

- this but in paper

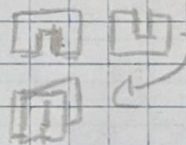
- origami crane ← why not

- Recut origami but keep it in 2D, user folds it themselves

- first card but more detailed cuts

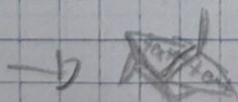
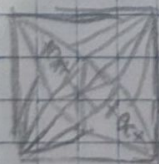
- silhouette from two sides but they have a slot so they can stack like a cheap video game asset

- not really 3D



Paper crane:

- keep name and contact info on wings (most surface area, too many folds otherwise)
- get crease lines from Google and cut along those, leave text where wings are supposed to go



Reflection:

I don't think the design funnel is really helpful. The initial brainstorming is nice so I'm not too caught up in one idea but I can't really develop them using the design funnel. Too many are dropped too quickly to be useful

Reactions (to business card):

* Persons 1-3 were interviewed
in the tech lab

Person 1: (from tech lab)

- "wow"
- *immediately accidentally drops it
- "how did you make the text line up with the folds"
- "very sturdy"
- "cool"

Person 2: (from tech lab)

- "so did you laser cut first, then fold it"
- "so innovative"
- "where are the engravings"
- "good insight" (regarding exporting software)
- "good idea"

Person 3: (from tech lab)

- "so was there a grade for creativity"
- "lucky fun"
- "is that made out of cardstock"
- "cooler than those thin metal cards"
- "must have been a pain to fold"

Person 4:

- Feneron Wang
- add feathers
- more detail
- 3D is creative!!!
- Add something under the wings

Person 5: Olivia Kim

- move letters so it isn't folded
- very cute
- I like the 3D
- cute that it brings back elementary memories
- give it eyes.

Business Card

https://drive.google.com/drive/folders/1D8Ka9IZsst7BV-g1darz4eS5UFz1efVX?usp=drive_link



I verify Fengran Wang demonstrated the Arduino part A assignments to me and it functioned according to the project requirements. Following Emory's Honor code, I certify that my verification is in keeping with the expectations to truthfully and faithfully represent my experience.

Norman¹²⁰ Oct. 29, 2024

LED Assignment

https://drive.google.com/drive/folders/1Dohx4eIZUBSK1FyN6lN8EpFojZSl6g5V?usp=drive_link



Resistor Assignment

https://drive.google.com/drive/folders/1HiPcC2bJ2ARyBgCOIm0AnsN-9sAIFKrq?usp=drive_link



I verify Shaniah Reece demonstrated the Button Assignment to me and it functioned according to the project requirements. Following Emory's Honor Code, I certify that my verification is in keeping with the expectation to truthfully and faithfully represent my experience.

Kenneth

Sept. 30, 2024

Button Assignment

https://drive.google.com/drive/folders/1jYphaTsdmen2kfu0_szmWvXlNmXsUucM?usp=drive_link



(there was alot of reused code between these assignments)

Final Project Ideas (using 10x 10 method)

challenge: cheap, easily replicable electronics

1. watch with display and additional sensors on clothes for monitoring
2. exoskeleton arm with motors (??) printed chassis? (electrodes or buttons on arm for control)
3. powered gloves with motors to give feeling and resistance when interacting in VR
4. very small and cheap drone (??) printed chassis)
5. separate robotic arm that mimics a human arm (for use in dangerous(?) situations)
6. self centering mouse pad (motorized) don't have to move mouse back to center
7. wearable self retracting mask (like the ones you see in movies)
8. sensor to tell if you've left the stove on when the front door is opened
9. detect facial movement using electrodes then map onto a virtual avatar
10. put a small language model on an Arduino (no clue how that one goes wrong)

Selected Idea: exoskeleton arm (because it sounds cool)

Details: (kind of like several above ideas rolled into one)

1. more out of wood probably could see if I can get aluminum
2. inner layer will be lined with buttons to detect movement
3. joints will have potentiometers to detect movement
4. hands will be actuated by strings
5. joints will also have motors
6. will likely just be a frame to keep it light
7. inner layer may be separate
8. sensing component and actuator on different circuits
9. will probably only be for one arm
10. may need more than 1 Arduino

Elevator pitch

- focus: video games and entertainment/cultural applications
- will have little practical uses and side that one
- since it is only a small project
- use in VR to give tactical feedback
- cheap & open source
- cool play(?)

Robotics
to serve
the same
function
as the buttons

buttons, when pressed
it moves
an arm has
motors and
the motors
should
move with
it

vi:

What's wrong with VR today? Sure, you can see objects and interact with them but you can't feel them. It's especially a pain dropped to your face, I want more from my gaming experience. I want what has been promised in movies, books and ads alike. As such, I propose an exoskeleton arm that can sync the virtual and the real in a way that players can finally feel for themselves.

Feedback: (20 years older, business major)

- "why? why an 'exoskeleton' though?" → how do I say "It's cool profession"
- "isn't meta already doing this?" → not just for gaming

vii:

In this day and age, the line between what is virtual and what is real is already blurry. I, however, want to blur it some more. I want to make a mechanical exoskeleton that can serve both as a way to interface with VR and a way to bring VR to life. It will be able to both provide tactical feedback in the virtual world and bring the beloved characters to life through cosplay.

Feedback: (30+ years older, doctor + investor)

- "who would use this?" (serious question, not mocking)
- "how strong will it actually be?" → not strong, but cheap
- "why would you need this for cosplay?" → you can use it for anything

viii:

In this day and age, the line between what is virtual and what is real is already blurry. I, however, want to blur it some more. Virtual Reality already is already a massive market for games yet it only includes two of four senses. I want to up that to three, with an Arduino powered exoskeleton, we can bring tactile sensitivity and feedback to VR. Not only that, with how cheap and accessible the open-source Arduino is, it can have a wide range of applications besides. Everything from being used in cosplay to medical applications.

I verify Sam Liu demonstrated the speaker assignments to me and it functioned according to the project requirements. Following Emory's honor code, I certify that my verification is in keeping with the expectation to truthfully and faithfully represent my experience.

Kenn^{DD} Oct. 4, 2024

Observations (for midterm)

- surprised by the scrolling display (but said it was nice)
- surprised that buttons can be held down
- comment that walls and player should be more distinct
- path should be wider
- animations are nice but can be too flashy to interpret

Link to midterm:

Link to research (for finals)

OK Matrix (Midterm)

https://drive.google.com/drive/folders/135CGWCATQf60meZ8QLR4NDt2PI-FRBV?usp=drive_link



Made a 2D maze

There is a separate controller module as well for easier control

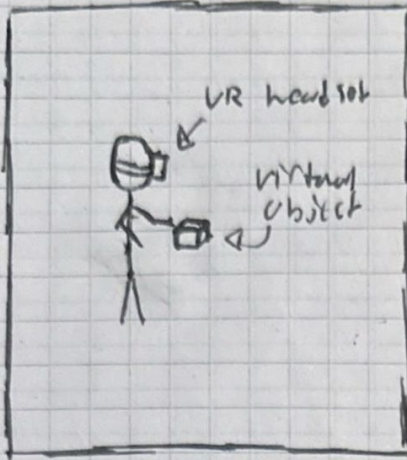
I also made a purely software markup in Processing to test the maze generation and logic

Literature Review

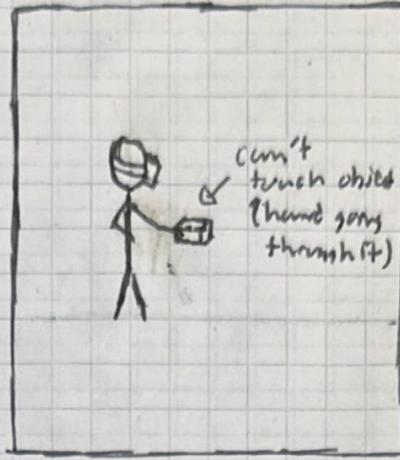
https://docs.google.com/document/d/15_wokSE6acw43eYB9L57uyNVGxmTy9ggwlvEKF3SDhY/edit?usp=drive_link



I am not in possession of enough props for this so here are some badly drawn stick figures instead

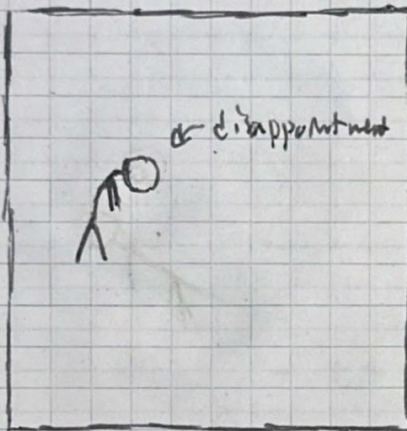


Playing VR, wants to touch object

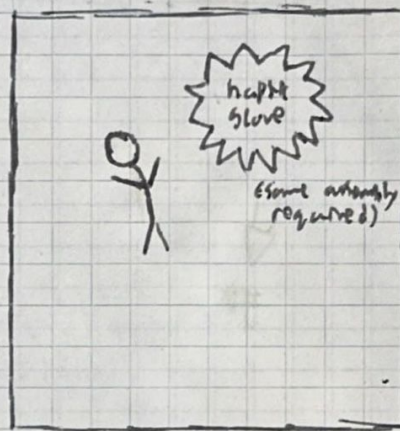


cannot touch object as it is virtual

why am I incapable of art

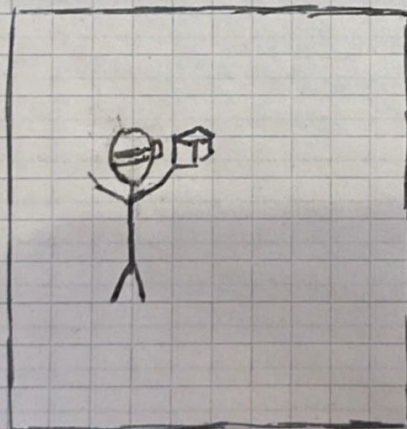


find that the object could not be touched



user discovers and assembles a haptic glove

I can barely draw stick figures, I am not drawing a glove



VR object can now be touched

Emphasize that it is cheap that is the biggest edge, then that it is cheap and easy to assemble

Sound PWM

https://drive.google.com/drive/folders/19ITWwebnRaF27vBcJXfc_h3z6QYBzkqB?usp=drive_link



Bonus: It plays Bad Apple

You can't have a display and a speaker and NOT try to make it play Bad Apple.

(I did in conjunction with OK (robot)
(ran out of memory on the Arduino
for the full song/display so it
only has a part of the song)

Servo Assignment

https://drive.google.com/drive/folders/10TX3luroDqNfqZ_H1tP1u3xKe7POkb6?usp=drive_link



Love me some servos

I feel like I bought too many
but given how often I burn
out motors in Robotics, maybe
that's a good thing.

3D Printing:

https://drive.google.com/drive/folders/1zHJtWIL1bmYpIzM2453ejufHpIYCh3H?usp=drive_link



Ran out of filament more times
than I can count, last calibration
job ended with me hand feeding
the filament because it was
no longer long enough to stretch
from the extruder to the feeder.
(I have a graveyard of failed prints).

Pressfit Cube:

https://drive.google.com/drive/folders/1F_EJtnRvo6Xkt3BUkbMWB1_EMABYmXel?usp=drive_link



The final cut was done at the last minute and I didn't have time to do another one so one of the panels are a bit loose. Not bad though considering I freehanded everything in lightburn for the actual cut.

(I also made a much smaller scaled down wooden cube as a test and it is so adorable I have formed an emotional attachment to it)

Soldering:

https://drive.google.com/drive/folders/1HJQBduBWEuj0aYCFVKChLqHknSNylmH?usp=drive_link



hmm... that's color of fumes.

Build Your Own Arduino

https://drive.google.com/drive/folders/128h8NAr7iMpwRpllwO41kUmyXuaxOeW?usp=drive_link



Photos of breadboard Arduino are included, as well as all 3 iterations of the PCB design. PCB has not been soldered yet though.

final project (actual build)

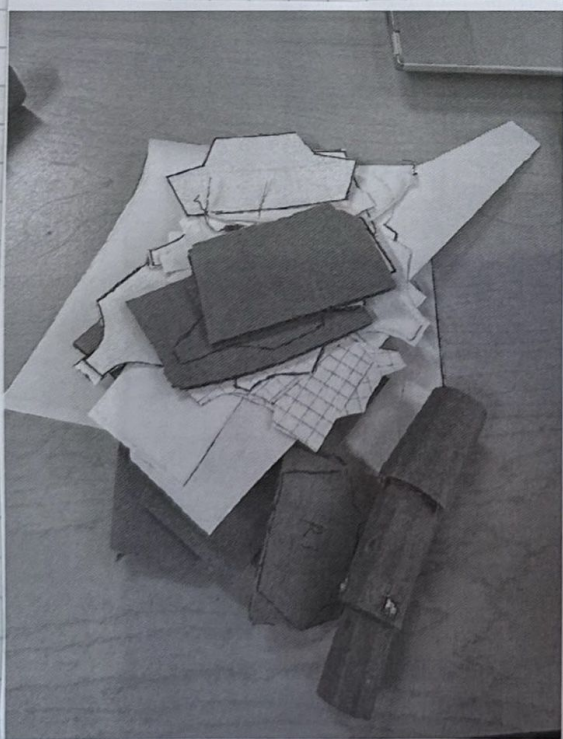
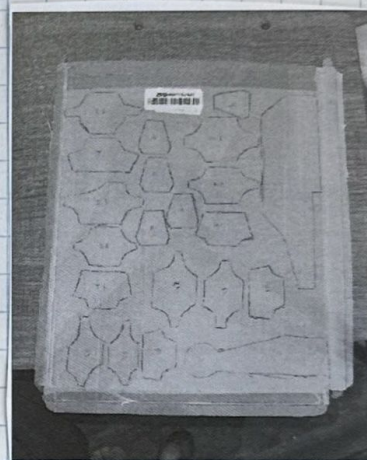
Final Project

https://drive.google.com/drive/folders/1l0t5gF7ZQ_5wH2ua_zxJZ7dvHK7c_Yu7?usp=drive_link



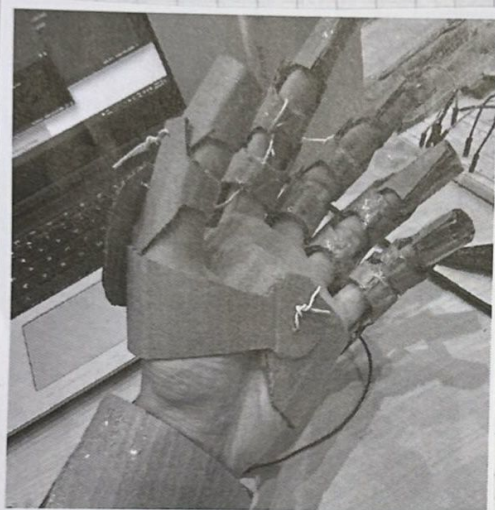
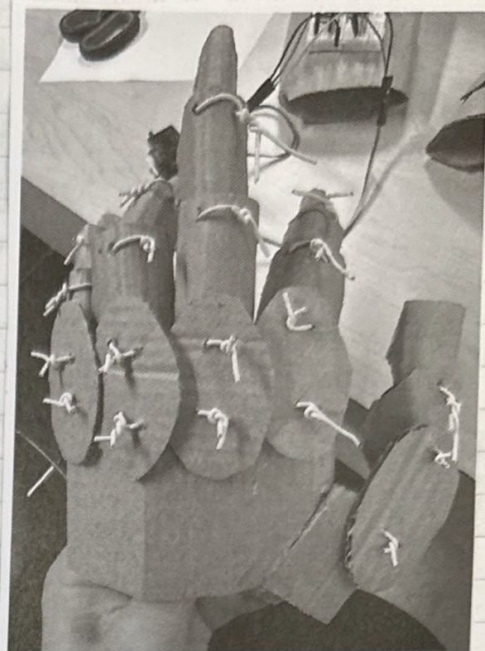
(Photos and code here)
(link of video too)

The entire premise of the project is to make something cheap and easy to build. Not everyone has access to a 3D printer and the myriad of sensors and actuators that other hobbyist projects have. So, I will be making mine out of nothing but cardboard and a handful of servo motors.



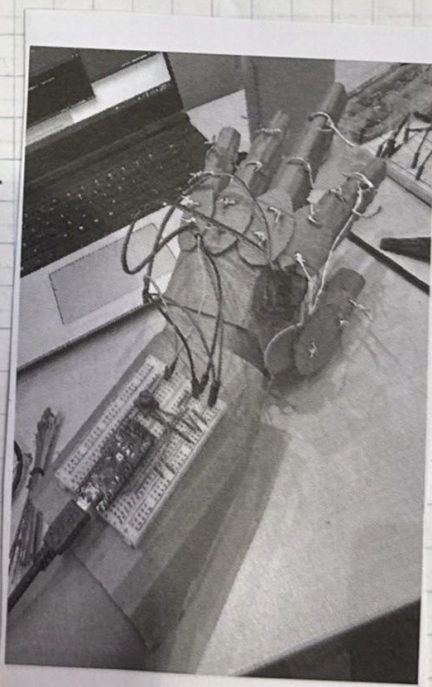
First attempt ended in failure, the cardboard I used was too soft and the cut itself was far too small. I now need a rec. Extra finger segments did come in handy for testing different joints though.

Second attempt used copper wire as fasteners on the joints, individual segments are held together by tape. This provides excellent mobility and is actually surprisingly sound structurally speaking. Does not seem strong enough for the motor to pull through.

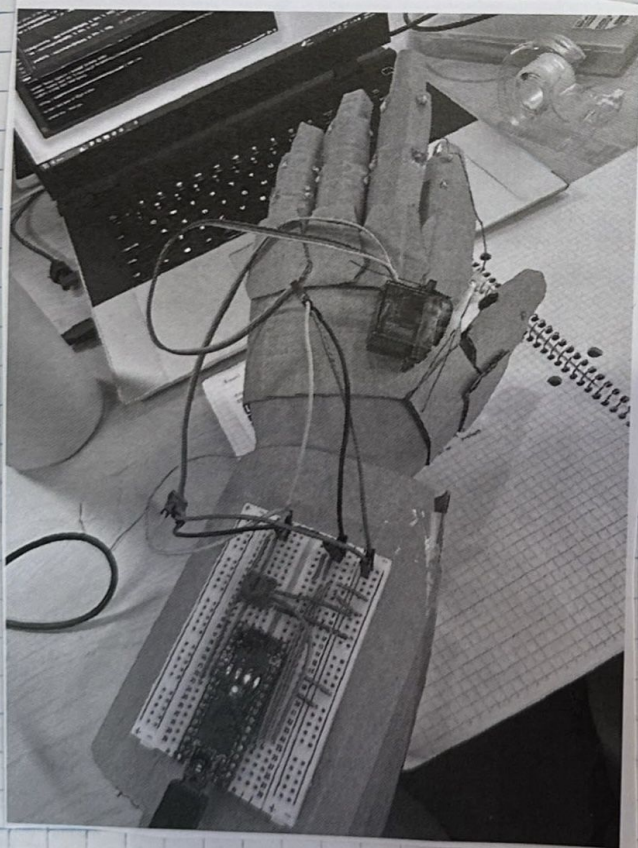


Added a forearm brace as a secondary point for electronics.

Added electronics (bread board and servo) and it actually works. The servo current draw (detected via a shunt to the analog pins) spikes whenever it is pushed. Depending on which servos current spikes, I can then determine which way the two of them are being pulled. I can then determine whether or not to allow them to push forward. Unfortunately, using this system, I need 2 servos and 2 corresponding analog pins per each finger and since I have 5 fingers and the Arduino has 8 analog ports, I cannot have every finger actuated. I'll just do one, for now.



Attempt 1: Re-did entire glove body in an attempt to make it more rigid so curves are more effective. This time I am using copper wire (with insulation) as joints but this time I superglued them in place. Scrap joints/finger segments from the previous two attempts were helpful in practicing making the joints.



Finished product: It's done. This iteration is far more rigid and holds its shape better than the previous one. I also replaced the wire running between the motor and the finger ends with threads. This provided better tension as the wire was too rigid (broke) but the index finger can now be moved with just the sensors. Unfortunately, the cardboard flexes far too much for the sensors to work as that requires non-instantaneous force to be applied to the sensors to work, with some time and a proper mount instead of double sided tape in attempt 3 and scratch tape in attempt 2. I think this does have potential.

Interview!

Goal: establish whether this project has achieved the goal of making something anyone can build and is more accessible than mainstream alternatives.

Questions:

- Can you tell what this even is?
- Do you think you would be able to make this your self?
- Do you think you would be able to easily acquire the materials to build this?
- Compared to other glasses on the market, how do you think this compares in terms of price and accessibility?
- Questions? (Do demo)

Group Interview Recording

(unintentional paper)

https://drive.google.com/file/d/1FG2chSoqDJxWH8_ifr3V9_iTCd5tzZGW/view?usp=drive_link



Went pretty well, main concern seemed to be getting this into actually working wires the secondary concern of getting this to fit a variety of head sizes (something that came up in research as well). All interviewees seemed to feel confident that they can assemble this themselves though so that is a win.

Final Project Code Github

<https://github.com/Kevinqu2004/CS-584-Final-Project-Code/tree/main>



Copy of file one in Google Drive.